

GOSFORD HIGH SCHOOL



Year 12

2011

HSC

MATHEMATICS

Assessment Task #2

Time Allowed: 90 minutes+5 minutes reading time

Instructions:

- Start each question on a new sheet of paper.
- Attempt questions 1-5.
- Board approved calculators may be used.
- Write using black or blue pen.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.

QUESTION 1 (12 Marks) Start a new sheet of paper.

MARKS

- a) The gradient of a curve is given by $\frac{dy}{dx} = 3x^2 - 12$
- | | |
|--|---|
| (i) find $\frac{d^2y}{dx^2}$ | 1 |
| (ii) if the curve passes through (1,-2), find the equation of the curve | 2 |
| (iii) find the values of x for which the curve both increases and is concave downwards | 3 |
| (iv) find any point(s) of inflexion | 2 |
| (v) sketch the curve labeling all critical points (do NOT include x - axis intercepts) | 2 |
| b) Find exact values of x for which the gradient of the curve $y = 2x(x + 3)^2$ is 14. | 2 |

QUESTION 2 (12 Marks) Start a new sheet of paper.

- | | |
|--|---|
| a) Find the primitive function of $x^{-2} - 2$ | 2 |
| b) Find the indefinite integral of $\int (6x + 7)^3 dx$ | 2 |
| c) Evaluate (i) $\int_{-1}^2 (x^3 + x - 5) dx$ | 3 |
| Find (ii) $\int \frac{dx}{\sqrt{9-2x}}$ | 2 |
| d) Find the area bounded by the curve $y = \sqrt{x}$, the y -axis and the lines $y = 2$ and $y = 3$. | 3 |

QUESTION 3 (12 Marks) Start a new sheet of paper.

MARKS

- a) Evaluate $\log_2 32$ 1
- b) Solve $2^x = 1000$ to 3 significant figures 1
- c) Sketch neatly, and stating Domain and Range, the graph of $y = \ln x$ 2
- d) Find $\frac{dy}{dx}$ if $y = \ln \frac{2x+5}{3-x}$ 2
- e) Evaluate $\int_1^e x^2 + \frac{2}{x} dx$ 2
- f) (i) Sketch the curve $y = \frac{1}{3-x}$, showing any critical information 2
- (ii) Find the area enclosed by: $y = \frac{1}{3-x}$, the x -axis and the lines $x = 1$ and $x = 2$. 2

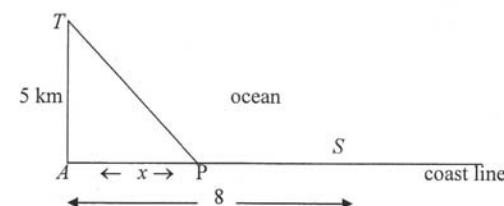
QUESTION 4 (12 Marks) Start a new sheet of paper.

- a) Sketch neatly, showing any critical points / values, $y = e^x - 1$ 2
- b) Differentiate $y = e^{\sqrt{x}}$ 2
- c) Find the stationary point of $y = xe^{-x}$ and determine its nature. 3
- d) (i) Find the area bounded by the curve $y = e^{2x}$, the x -axis and the lines $x = 1$ and $x = 3$ 2
- (ii) Rotate this area around the x -axis and evaluate the exact volume of the solid formed 3

QUESTION 5 (12 Marks) Start a new sheet of paper.

MARKS

- a) A natural gas pipe line is to be built connecting a coastal city S to an offshore island T which is 5 km from the closest coastline point A . The distance between A and the city S is 8 km. The pipeline is to be run from S to a point P then underwater to T . The cost of laying the pipeline is \$75 000 per km on land and \$100 000 per km under water.

Let $AP = x$

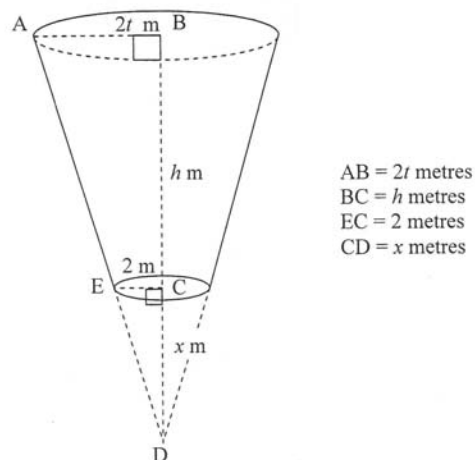
- (i) show that the length of the pipeline (l) is $\sqrt{x^2 + 25} + (8 - x)$ 2
- (ii) find an expression for the cost C of building the pipe line. 2
- (iii) find where P should be located to minimise the cost of the pipeline 2

Question 5 continued over the page

QUESTION 5 (b) Continued

MARKS

- b) A truncated cone is to be used as a part of a hopper for a grain harvester. It has a total height of h metres. The top radius is to be t times greater than the bottom radius which is 2 metres.



- (i) If x is the height of the removed section of the original cone, show using similar triangles that $x = \frac{h}{t-1}$ 2

- (ii) Show that the volume of the truncated cone is given by $V = \left(\frac{4\pi h}{3}\right)(t^2 + t + 1)$ 2

- (iii) If the upper radius plus the lower radius plus the height of the truncated cone must total 12 metres, calculate the maximum volume of the hopper. 3

END OF TEST ☺

Q1 a) $\frac{dy}{dx} = 3x^2 - 12$

i) $\frac{d^2y}{dx^2} = 6x$

ii) $y = x^3 - 12x + C$

$(1, -2)$ lies on this

$-2 = 1 - 12 + C$

$9 = C$

$\therefore y = x^3 - 12x + 9$

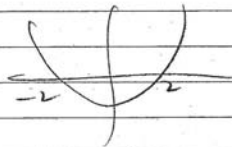
iii) Increasing $y' > 0$

$3x^2 - 12 > 0$

$x^2 - 4 > 0$

$(x-2)(x+2) > 0$

$x > 2 \vee x < -2$



Concave down

$y'' < 0$

$6x < 0$

$x < 0$

$\therefore x < -2$

3

iv) $6x = 0$
 $x = 0$

$y = 9$

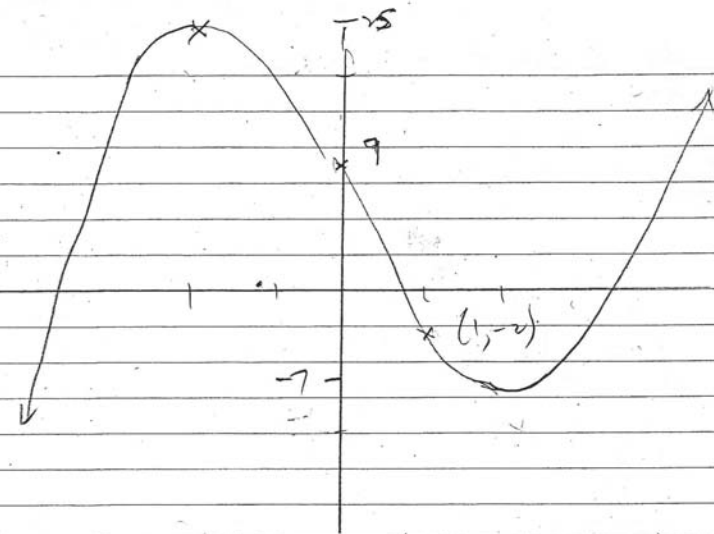
$(0, 9)$

check concavity change

x	0	0	0	+
y''	-	-	0	+

Concavity changes
down to up.

$(0, 9)$ is a pt of inflexion.



Stationary pts $y' = 0$

$3x^2 - 12 = 0$

$x^2 = 4$

$x = \pm 2$

$(-2, 25)$

$(2, -5)$

Nature

$x = 2$
 $y'' +$
⌒

$x = -2$
 $y'' -$
⌒

b)

$y = 2x(x+3)^2$

$y = 2x(x^2 + 6x + 9)$

$y = 2x^3 + 12x^2 + 18x$

$y' = 6x^2 + 24x + 18$

$14 = 6x^2 + 24x + 18$

$0 = 6x^2 + 24x + 4$

$0 = 3x^2 + 12x + 2$

$x = \frac{-12 \pm \sqrt{144 - 4 \times 3 \times 2}}{6}$

$x = \frac{-12 \pm \sqrt{120}}{6}$

$x = \frac{-6 \pm \sqrt{30}}{3}$

Question 2

$$a) \quad \frac{x^{-1}}{-1} - 2x + c$$

$$= -\frac{1}{x} - 2x + c$$

$$b) \quad \int (6x+7)^3 dx = \frac{(6x+7)^4}{4 \times 6} + c$$

$$= \frac{(6x+7)^4}{24} + c$$

$$c) \quad i) \quad \int_{-1}^2 x^3 + x - 5 \, dx$$

$$\left[\frac{x^4}{4} + \frac{x^2}{2} - 5x \right]_{-1}^2$$

$$= \left(\frac{16}{4} + \frac{4}{2} - 10 \right) - \left(\frac{1}{4} + \frac{1}{2} - 5 \right)$$

$$= 4 + 2 - 10 - \frac{1}{4} - \frac{1}{2} + 5$$

$$= -9\frac{3}{4}$$

$$ii) \quad \int (9-2x)^{-\frac{1}{2}} dx$$

$$= \frac{2(9-2x)^{\frac{1}{2}}}{-\frac{1}{2}} + c$$

$$= -\sqrt{9-2x} + c$$

$$d) \quad y = \sqrt{x}$$

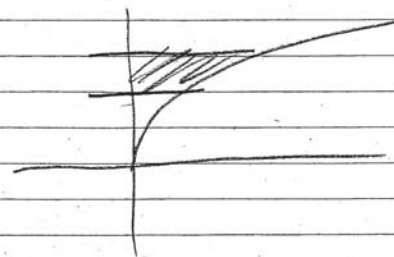
$$A = \int_0^4 dx \, dy$$

$$A = \int_2^3 y^2 dy$$

$$= \left[\frac{y^3}{3} \right]_2^3$$

$$= \frac{27}{3} - \frac{8}{3}$$

$$= 6\frac{1}{3} \, u^2$$



Question 3

a) $\log_2 32 = x$

$$2^x = 32$$

$$x = 5$$

$$\log_2 32 = 5$$

(1)

b) $2^x = 1000$

$$x \ln 2 = \ln 1000$$

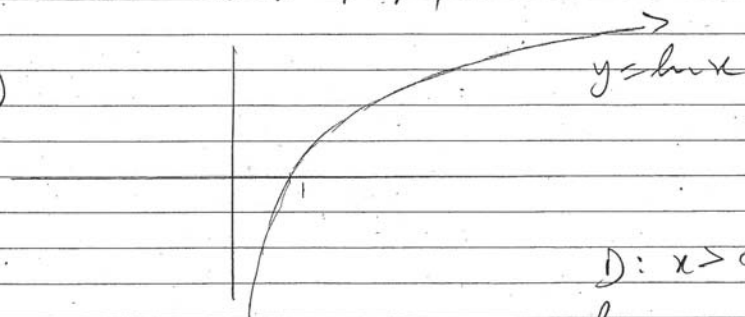
$$x = \frac{\ln 1000}{\ln 2}$$

$$= 9.96578$$

$$x \approx 9.97$$

(1)

c)



$$D: x > 0$$

Range: all real y

d) $y = \ln \frac{2x+5}{3-x}$

$$y = \ln(2x+5) - \ln(3-x)$$

$$\frac{dy}{dx} = \frac{2}{2x+5} - \frac{-1}{3-x}$$

$$= \frac{2}{2x+5} + \frac{1}{3-x}$$

OK $\frac{f'(x)}{f(x)}$

$$\frac{2(3-x) + 2x+5}{(2x+5)(3-x)}$$

$$6 - 2x + 2x + 5$$

$$\frac{11}{(2x+5)(3-x)}$$

$$= \frac{11}{(2x+5)(3-x)}$$

$$\begin{aligned}
 e) \quad & \int_1^e x^2 + \frac{2}{x} dx \\
 & = \left[\frac{x^3}{3} + 2 \ln x \right]_1^e \\
 & = \frac{e^3}{3} + 2 \ln e - \left(\frac{1}{3} + 2 \ln 1 \right) \\
 & = \frac{e^3}{3} + 2 - \frac{1}{3} - 0 \\
 & = \frac{e^3}{3} + \frac{5}{3} = \frac{e^3 + 5}{3}
 \end{aligned}$$

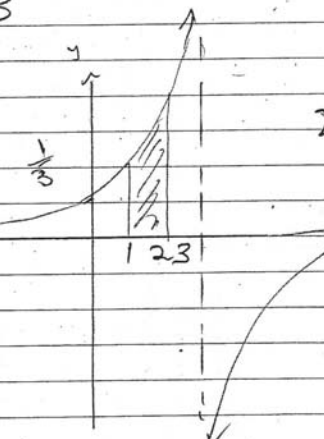
f) i) $y = \frac{-1}{x-3}$

$A = \int_1^2 \frac{-1}{3-x} dx$

$= \left[-\ln(3-x) \right]_1^2$

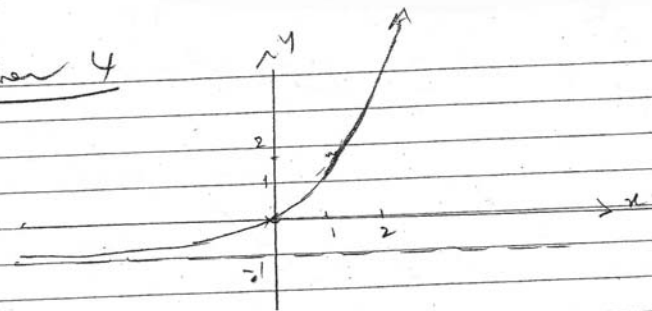
$= -\ln 1 + \ln 2$

$= \ln 2$



Question 4

a)



b)

$$y = e^{x^2}$$

$$y' = e^{\sqrt{x}} \cdot \frac{1}{2} x^{-1/2}$$

$$y' = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

c)

$$y = x e^{-x}$$

$$y' = x \cdot -e^{-x} + e^{-x}$$

$$y' = e^{-x} (1-x)$$

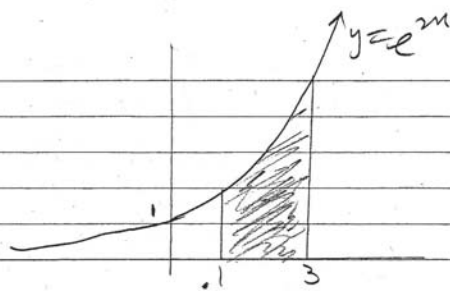
Stat at $x=1$ $y = \frac{1}{e}$

Nature

x	1	1	1
y'	+	0	-

$(1, \frac{1}{e})$ is a maximum turning point

d) i)



$$A = \int_1^3 e^{2x} dx$$

$$= \left[\frac{e^{2x}}{2} \right]_1^3 = \frac{e^6}{2} - \frac{e^2}{2} \text{ m}^2$$

$$\text{ii) } V = \pi \int_2^3 y^2 dx$$

$$y = (e^{2x})^2 = e^{4x}$$

$$V = \pi \int_2^3 e^{4x} dx$$

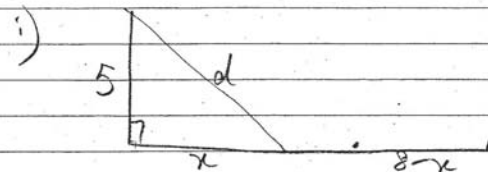
$$= \pi \left[\frac{e^{4x}}{4} \right]_2^3$$

$$= \pi \frac{e^{12} - e^8}{4}$$

$$= \frac{\pi}{4} e^8 (e^4 - 1) \text{ m}^3$$

Question 5

a) £ 75 000 in land £ 100 000 under the



Using Pythagoras $d^2 = 5^2 + x^2$
 $d = \sqrt{25 + x^2}$

$$\therefore \text{Length} = \sqrt{25 + x^2} + (8 - x)$$

$$\text{ii) } C = 100000 \sqrt{25 + x^2} + 75000(8 - x)$$

$$C = 25000 \{ 4 \sqrt{25 + x^2} + 3(8 - x) \}$$

$$\text{iii) } \text{For minimum cost } \frac{dC}{dx} = 0$$

$$\frac{dC}{dx} = 25000 \left\{ 4 \times \frac{1}{2} (25 + x^2)^{-\frac{1}{2}} \cdot 2x + -3 \right\}$$

$$0 = \frac{4x}{\sqrt{25 + x^2}} - 3$$

$$3 \sqrt{25 + x^2} = 4x$$

$$9(25 + x^2) = 16x^2$$

$$225 + 9x^2 = 16x^2$$

$$225 = 7x^2$$

$$x^2 = \frac{225}{7}$$

$$x = 5.669 \text{ km}$$

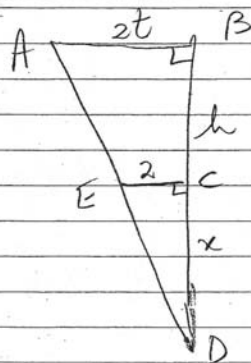
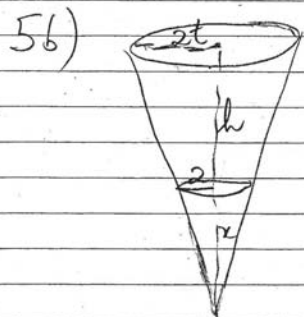
$$x \doteq 5.67 \text{ km from A}$$

Show cost is in fact minimum

x	5	5.7	6
C'	-17	0	0.07

- 0 +

$\therefore$ cost is minimum



In $\triangle EDC$, $\triangle ADB$

1. $\angle D$ is common
2. $\angle ECD = \angle ABD$ (both 90°)

$\therefore \triangle EDC \parallel \triangle ABD$ (angles test)

$$\therefore \frac{x}{x+h} = \frac{2}{2t} = \frac{1}{t}$$

$$tx = x+h$$

$$tx - x = h$$

$$x(x-1) = h$$

$$x = \frac{h}{t-1} \quad \text{as required}$$

$V = \text{Large} - \text{small cone}$

$$= \frac{1}{3} \pi (2t)^2 (x+h) - \frac{1}{3} \pi (2)^2 x$$

$$= \frac{\pi}{3} \{ 4t^2 x + 4t^2 h - 4x \}$$

$$= \frac{4\pi}{3} \{ x(t^2 - 1) + t^2 h \}$$

$$= \frac{4\pi}{3} \left\{ \frac{h}{t-1} (t-1)(t+1) + t^2 h \right\}$$

$$= \frac{4\pi}{3} \{ h(t+1) + t^2 h \}$$

$$= \frac{4\pi h}{3} \{ t+1 + t^2 \} \quad \text{as required}$$

iii) $2 + 2t + h = 12$

$$h = 10 - 2t$$

$$V = \frac{8\pi}{3} (5-t)(t+1+t^2)$$

$$= \frac{8\pi}{3} (5t + 5 + 5t^2 - t^2 - t - t^3)$$

$$= \frac{8\pi}{3} (4t^2 + 4t - t^3 + 5)$$

For max volume $\frac{dV}{dt} = 0$

$$\frac{dV}{dt} = \frac{8\pi}{3} \{ 8t + 4 - 3t^2 \}$$

$$3t^2 - 8t - 4 = 0$$

$$t = \frac{8 \pm \sqrt{64 + 48}}{6} = \frac{8 \pm \sqrt{112}}{6} = \frac{4 \pm 2\sqrt{7}}{3}$$

$$3.097 \quad \checkmark \quad -0.4305$$

$$\frac{d^2V}{dt^2} = \frac{8\pi}{3} \{ 8 - 6t \}$$

$$t = 3.097 \quad \frac{d^2V}{dt^2} = \frac{8\pi}{3} \{ 8 - 18.88 \}$$

negative

$\therefore$ Max volume when $t = 3.097$

$$t = -0.4305 \quad \frac{d^2V}{dt^2} = \frac{8\pi}{3} \{ 8 - -2.583 \}$$

pos $\therefore$ min volume

This Maximum volume is

$$V = \frac{8\pi}{3} \{ 4 \times 3.097^2 + 4 \times 3.097 - 3.097^3 + 5 \}$$

$$= \frac{8\pi}{3} (26.049)$$

$$V = 218.22 \text{ m}^3$$